

Nipun Chauhan

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SUMMARY

Data Analyst with hands-on experience in statistical modeling, data visualization, SQL, and applied machine learning; skilled at translating complex data into actionable business insights.

SKILLS

Languages & Frameworks: Python (Pandas, Matplotlib, PyTorch, Scikit-learn, NumPy), SQL, R

Databases: MS SQL, BigQuery, SQLite

Data Visualization: MS Excel, Power BI (DAX), KNIME

Modeling: Statistical Modeling, Machine Learning, Time Series Forecasting, Data Mining

WORK EXPERIENCE

American College of Emergency Physicians (ACEP)

June 2024 – Present

Data Science Intern

Irving, Texas

- Engineered a novel NLP pipeline by integrating Bayesian clustering with various BERT models to classify complex, unstructured medical text (no feasible rule-based alternative), and achieving 20% accuracy from 0% baseline.
- Optimized patient data workflows by analyzing 10M+ healthcare records, improving cross-system accuracy by 15%, and recommended the integration of the Claude 3.5 Sonnet API for better semantic alignment.
- Designed interactive Power BI dashboards to visualize data quality trends and model outputs.

PORTFOLIO PROJECTS

End-to-End Vendor Analysis for Profitability Optimization

May 2025 – June 2025

- Engineered an ETL pipeline transforming over 20 million records, reducing execution time by utilizing CTEs and unlocking analysis that identified a 72% bulk purchase savings opportunity.
- Uncovered critical profit opportunities using statistical models to validate that low-sales brands had significantly higher margins (41.6% vs. 31.2%), leading to a strategic pricing initiative for 198 target brands.
- Designed an interactive Power BI dashboard, using DAX to synthesize Python analysis into actionable KPIs.

Text-to-SQL LLM Application with Google Gemini Pro

July 2025 – Present

- Developed an end-to-end LLM application using Gemini that translates natural language questions into SQL queries, enabling non-technical users to extract insight from complex soccer databases.
- Engineered a schema-adaptive system that dynamically reads table structures to automatically inject the database schema into prompts, guiding the LLM to generate syntactically correct and context-aware queries.
- Built an interactive web (Streamlit) interface that executes generated queries and displays results in real-time.

Marketing Campaign Analysis: Facebook vs AdWords

June 2025 - July 2025

- Identified Facebook as the higher-converting platform using t-test ($p < .001$), revealing 96% higher daily conversions (11.74 vs. 5.98) than AdWords and driving a strategic budget reallocation to Facebook.
- Created a predictive model which enables accurate performance forecasting (80 clicks → 19 conversions), reducing customer acquisition costs through data-driven budget optimization.
- Discovered temporal conversion patterns revealing peak performance on Mondays/Tuesdays and cost efficiency in May/November (lowest Cost per conversion).

EDUCATION

The University of Texas at Dallas

August 2023 – May 2025

MS in Business Analytics and Artificial Intelligence (Data Science Track)

Kurukshetra University, Kurukshetra

August 2019 – July 2022

BS in Electronics, Physics & Mathematics